TECHNION | The Henry and Marilyn Taub Faculty of Computer Science

Selected Topics on Deep Learning over Structured Data

Course 02360028 (StructML)

Introduction

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Technion – Israel Institute of Technology

Winter 2026-27

The Course

- Paradigms for ML over **relational databases**
  - To this aim, ML over **tabular** and **graph** data
- Content:
  - 6 core modules
  - Tutorials: rehearsal, SQL, PyTorch, PyG, …
    - In general, the *practical & real-life side of the modules*
- Staff:
  - Lecturer: Prof. **Benny** Kimelfeld
  - TAs: Dr. **Shunit** Agmon (in charge), **Boaz** Berger
  - Graders: **Gal** Ishai, **Abed** el Rahman Badran
- English
- Slides and Spring-2026 recordings on the website
  - https://bennykcs.github.io/structml-course/

Requirements

- Two homework **assignments**
  - Total of 50% of the final grade
- Final **project**
  - 50% of the final grade
- Both assignments + project **mandatory**
- Each followed by an oral defense
    - Failure to defend the assignment will disqualify the assignment and the course participation
- All submissions are **individual** (not pairs)

Pedagogical Methodology & AI

- AI is allowed in assignments, coding/reporting
  - Even recommended
  - We will mark parts where AI usage is **not** recommended
- We require **full accountability**
  - As if you've written every character yourself
  - Full understanding of every line of code
  - Ability to explain & defend every design choice
  - Every mistake is yours
- Each assignment will include an **oral defense**
  - During the course hours
- Later lectures will include **self-learning**
  - From the slides and past recordings
  - The lectures will give a summary/complementation

Assumed Background

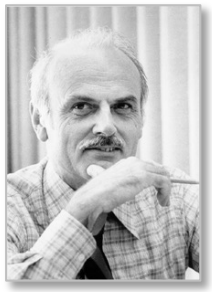
- Basic course on **machine learning**
  - Not necessarily *deep learning*
- **Databases** – an advantage
  - Not mandatory, mainly simple concepts used here, can be learned throughout the course

Structured ML

Context: Relational Databases

Relational DB: Codd's Vision (1970s)

- 1970: Codd invents the **relational database** model
  - Idea: interface via mathematical logic
    - Data = collection of relations, interconnected via keys
    - Questions via a query language over the schema
    - System translates queries into actual execution plans
  - Principle: separate logical from physical layers
  - Work done at IBM in San Jose, California
    - [E. F. Codd: *A Relational Model of Data for Large Shared Data Banks.* In Communications of the ACM 13(6): 377-387 (1970)]
    - [E. F. Codd: *Relational Completeness of Data Base Sublanguages.* IBM Research Report RJ987 (1972)]

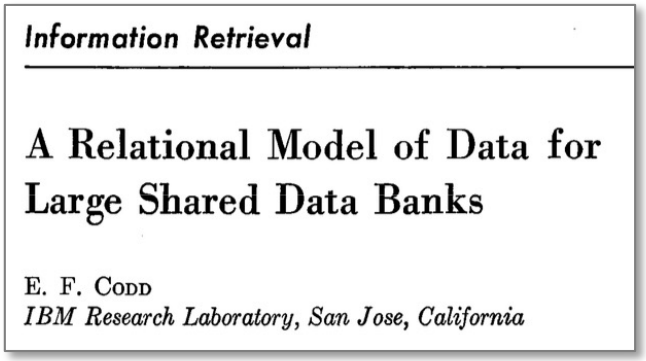


Edgar F. Codd

R	(*supplier*	*part*)	S	(*part*	*project*)
	1	1		1	1
	2	1		1	2
	2	2		2	1

FIG. 5. Two joinable relations

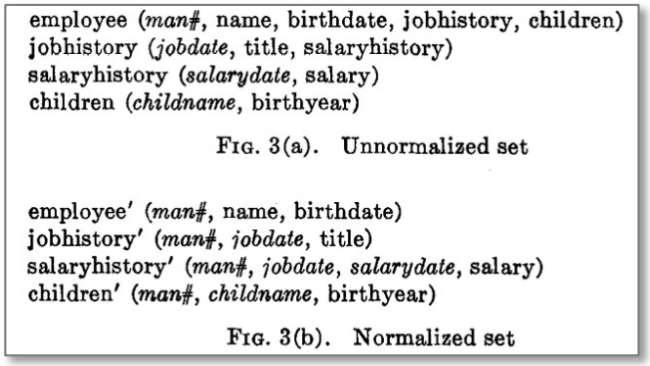
R*S	(*supplier*	*part*	*project*)
	1	1	1
	1	1	2
	2	1	1
	2	1	2
	2	2	1



Information Retrieval

A Relational Model of Data for Large Shared Data Banks

E. F. CODD
IBM Research Laboratory, San Jose, California



employee (*man#*, name, birthdate, jobhistory, children)
jobhistory (*jobdate*, title, salaryhistory)
salaryhistory (*salarydate*, salary)
children (*childname*, birthyear)

FIG. 3(a). Unnormalized set

employee′ (*man#*, name, birthdate)
jobhistory′ (*man#*, *jobdate*, title)
salaryhistory′ (*man#*, *jobdate*, *salarydate*, salary)
children′ (*man#*, *childname*, birthyear)

FIG. 3(b). Normalized set

Codd Catches On (1)

- 1973: Michael Stonebraker and Eugene Wong implement Codd's vision in **INGRES**
  - Commercialized in 1983
  - Evolved to **Postgres** (now PostgreSQL) in 1989



M. Stonebraker

INGRES



E. Wong

PostgreSQL

Codd Catches On (2)

- 1974: A group from the IBM San Jose lab implements Codd's vision in **System R**, which evolved to **DB2** in 1983
  - SQL initially developed at IBM by Donald D. Chamberlin and Raymond F. Boyce
    - [Chamberlin, Boyce: SEQUEL: A Structured English Query Language. SIGMOD Workshop, Vol. 1 1974: 249-264]
- 1977: Influenced by Codd, Larry Ellison founded Software Development Labs
  - Becomes *Relational Software* in 1979
  - Becomes *Oracle* Systems Corp (1982), named after its **Oracle** database product


R. F. Boyce (1947-1974)

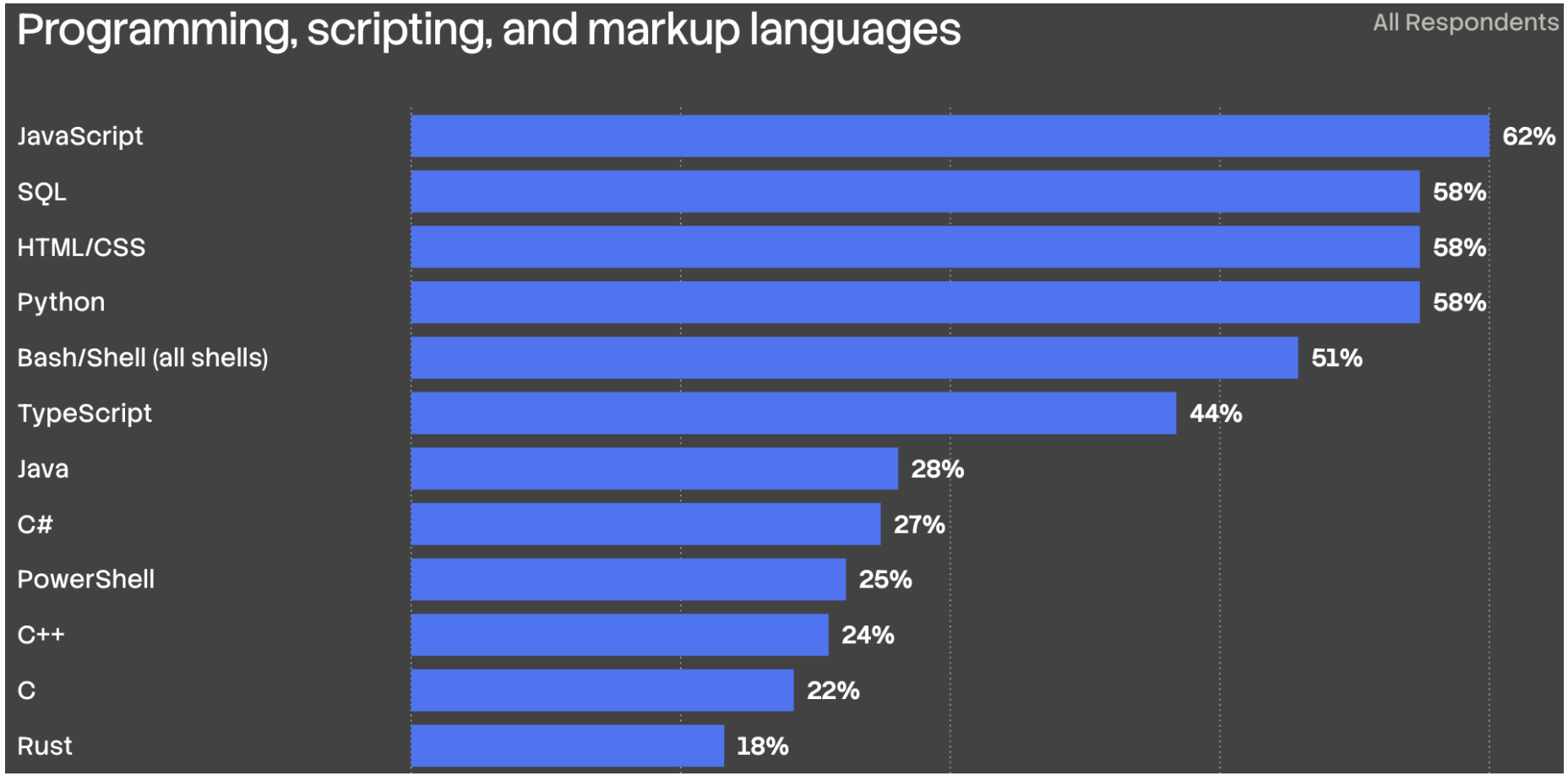

D. D. Chamberlin


J. Grey


L. Ellison


P. G. Selinger

DB Popularity in Programming

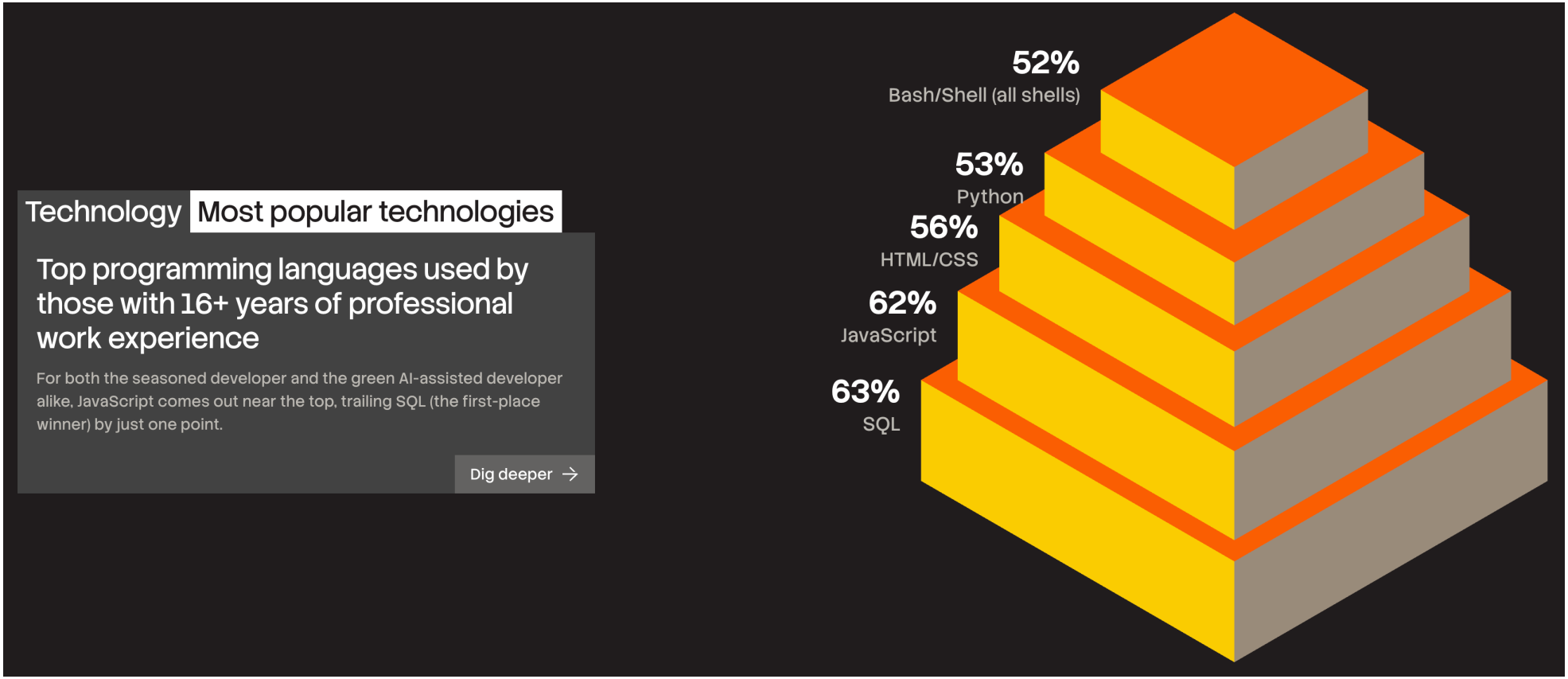


30,903 responses

Survey on Programming, Scripting, and Markup Languages

https://survey.stackoverflow.co/2026

DB Popularity in Programming

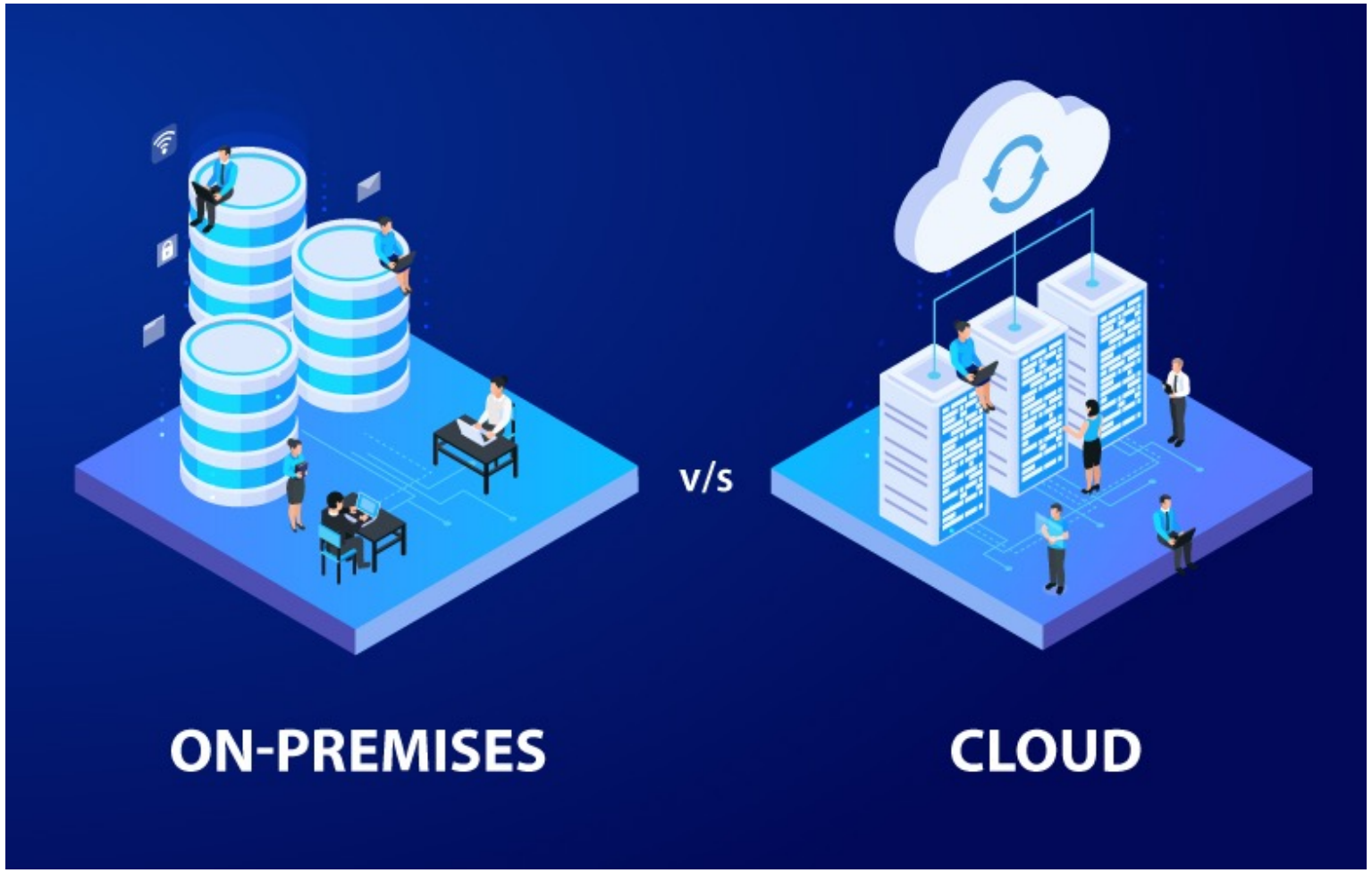


30,903 responses

Most popular technologies

https://survey.stackoverflow.co/2026

On-Permises vs. Cloud

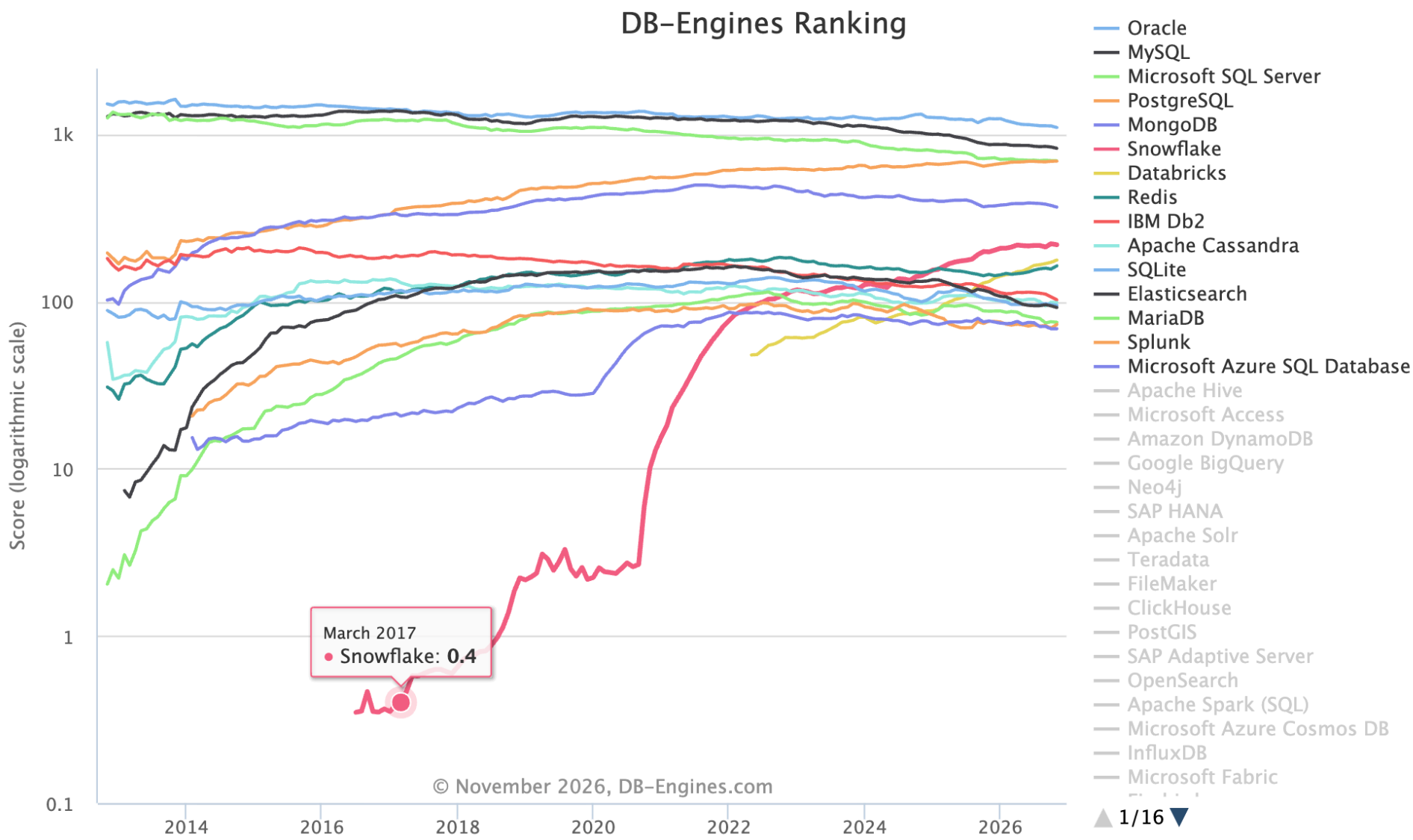


Database Ranking (Nov 2026)

Rank Nov 2026	Rank Oct 2026	Rank Nov 2025	DBMS	Database Model
1.	1.	1.	Oracle	Relational, Multi-model
2.	2.	2.	MySQL	Relational, Multi-model
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model
4.	4.	4.	PostgreSQL	Relational, Multi-model
5.	5.	5.	MongoDB	Document, Multi-model
6.	6.	6.	Snowflake	Relational
7.	7.	↑ 8.	Databricks	Multi-model
8.	8.	↓ 7.	Redis	Key-value, Multi-model
9.	9.	9.	IBM Db2	Relational, Multi-model
10.	10.	↑ 12.	Apache Cassandra	Wide column, Multi-model

https://db-engines.com/

Trends: Cloud (Snowflake, Azure) Rising



https://db-engines.com/

Structured ML

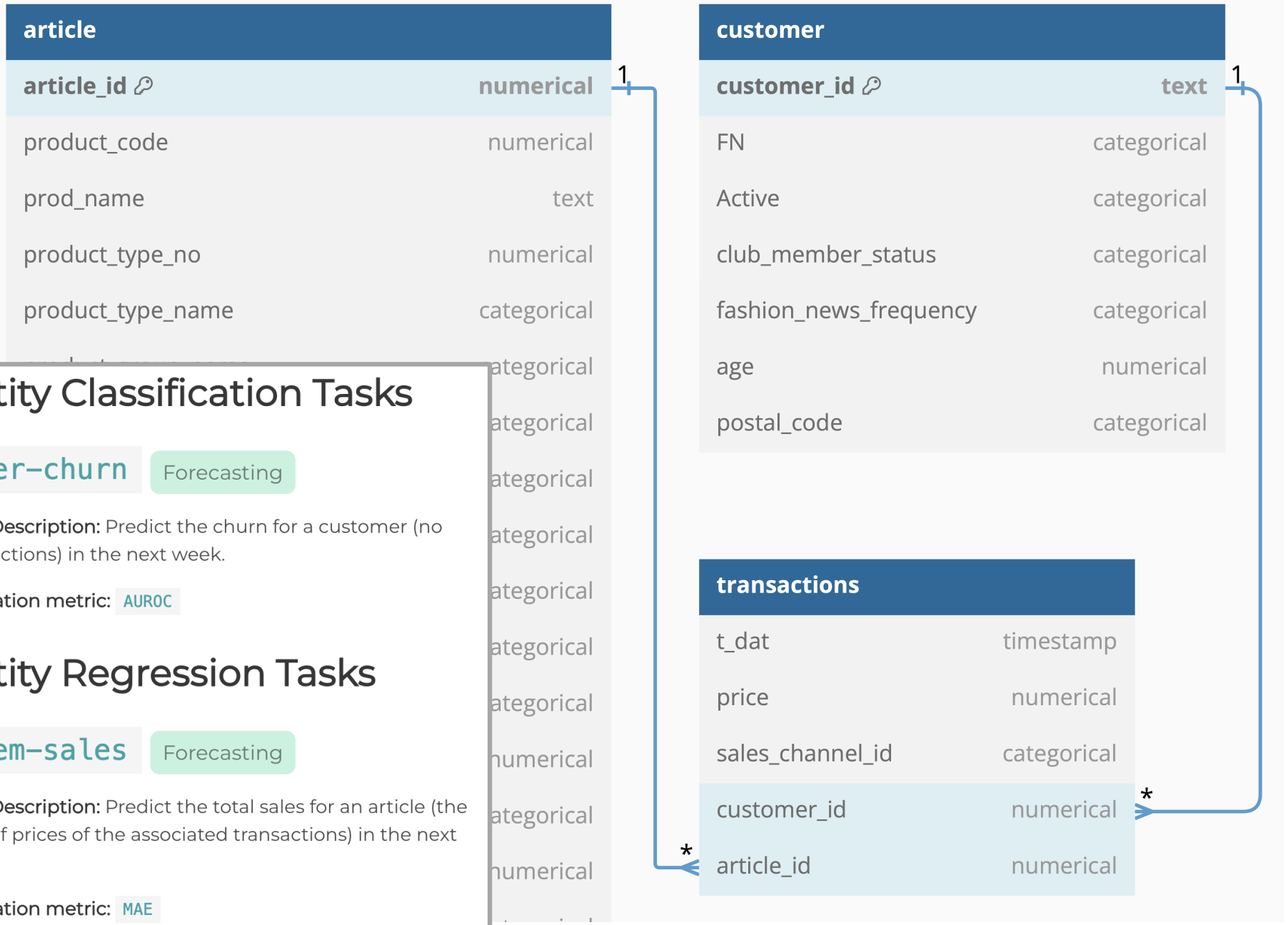
ML over Relational Databases

Databases: QL vs. ML

- Query languages:
  - Answer questions that we can translate into **exact** algorithms over the database
  - Designed for simple, high-level, **declarative** specification of information needs, without dealing with computation
- ML models:
  - Answer questions that involve uncertainty, such as prediction
  - We cannot phrase an exact query to answer our need
  - ... and yet, we believe that the *database information can greatly reduce the uncertainty*

Example: H&M DB

RELBENCH
RELATIONAL DEEP LEARNING BENCHMARK



Entity Classification Tasks

`user-churn` Forecasting

Task Description: Predict the churn for a customer (no transactions) in the next week.

Evaluation metric: `AUROC`

Entity Regression Tasks

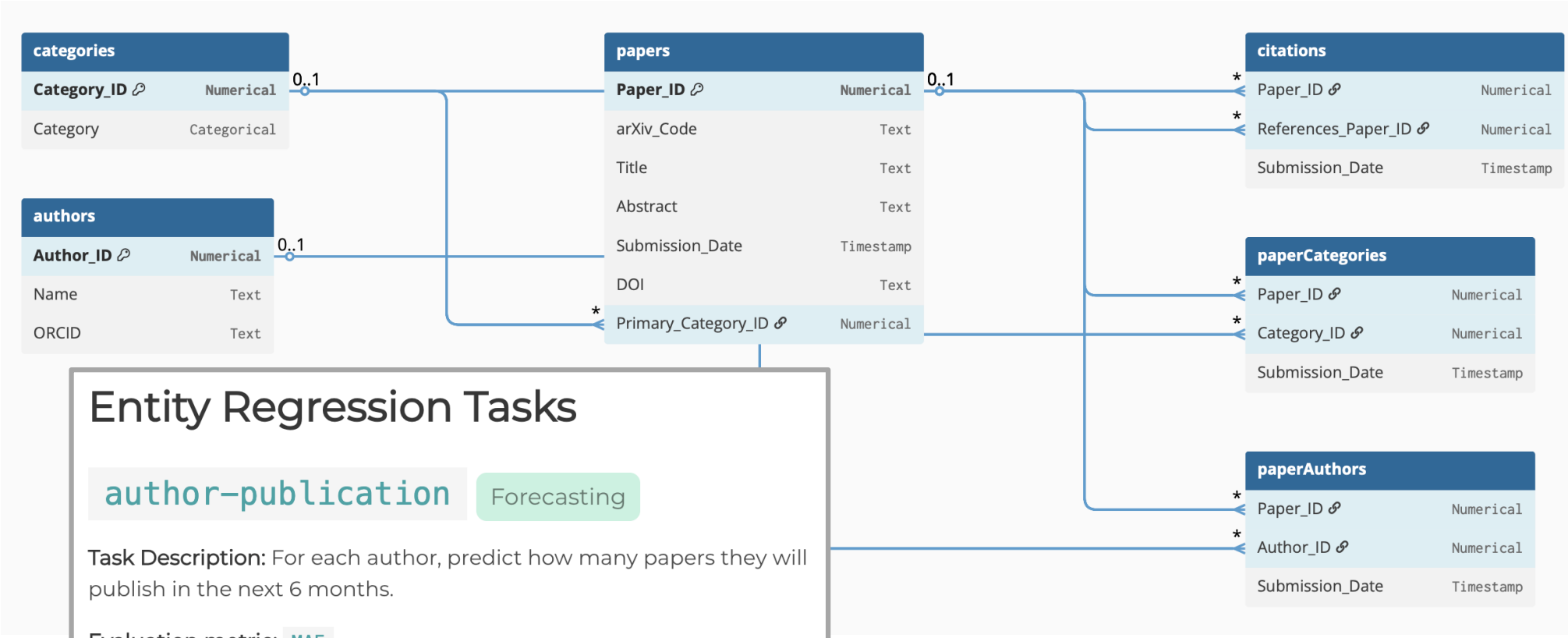
`item-sales` Forecasting

Task Description: Predict the total sales for an article (the sum of prices of the associated transactions) in the next week

Evaluation metric: `MAE`

Example: Arxiv (pub) DB

RELBENCH
RELATIONAL DEEP LEARNING BENCHMARK



Entity Regression Tasks

`author-publication` Forecasting

Task Description: For each author, predict how many papers they will publish in the next 6 months.

Evaluation metric: `MAE`

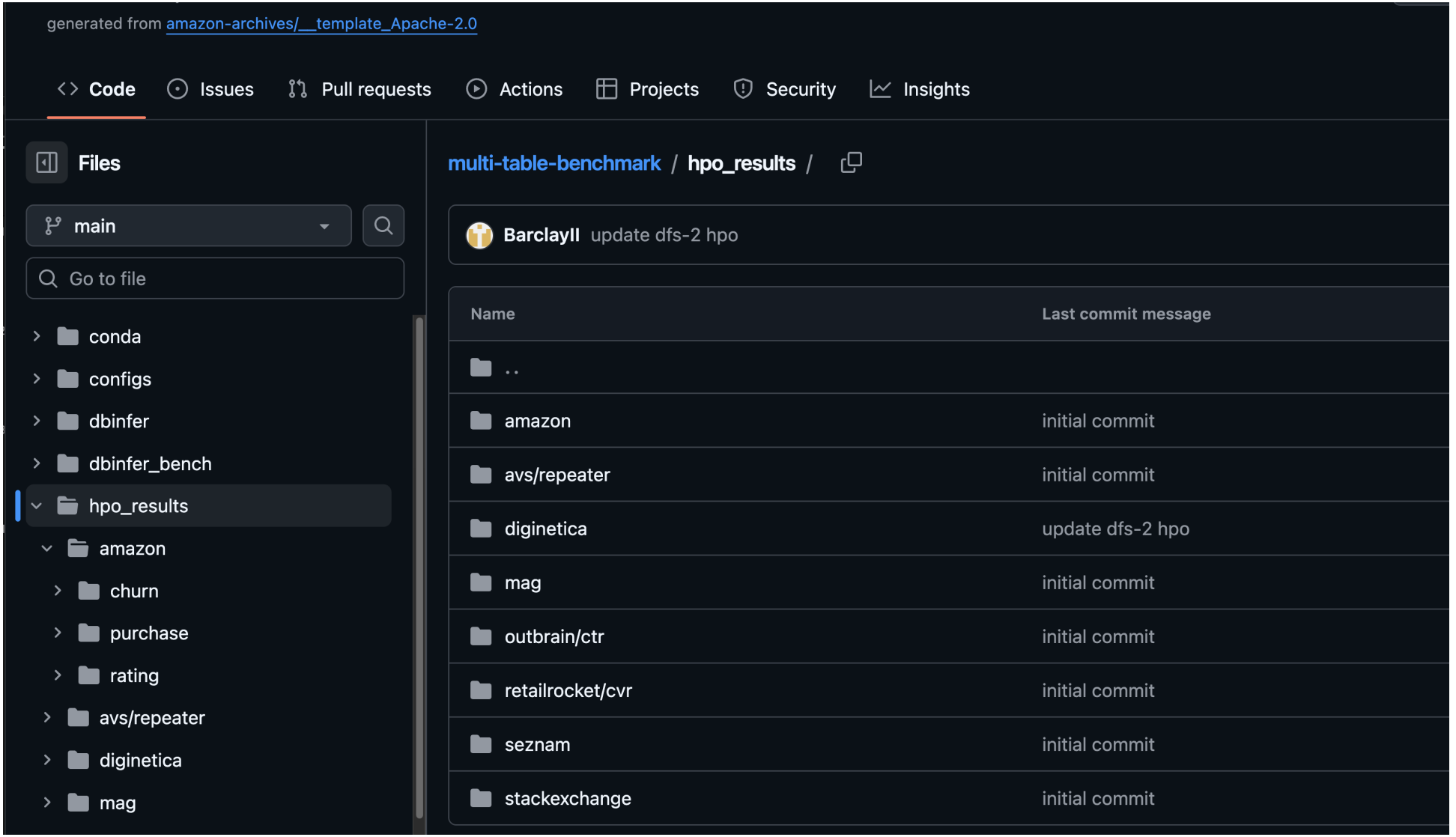
Link Prediction Tasks

`co-citation`

Task Description: For each paper, predict which other papers will be co-cited with it (i.e., cited in the same reference list) in the next 6 months.

Evaluation metric: `MAP`

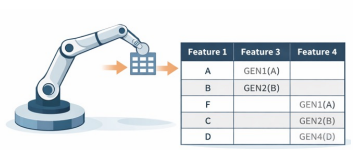
More from Amazon's Git

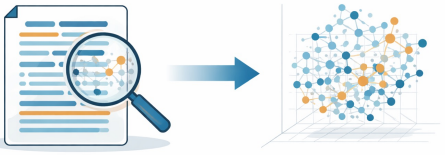


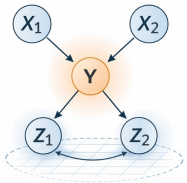
https://github.com/awslabs/multi-table-benchmark/tree/main/hpo_results

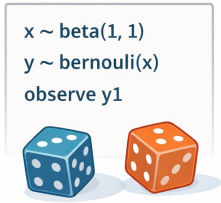
ML Paradigms: General & DB-Case

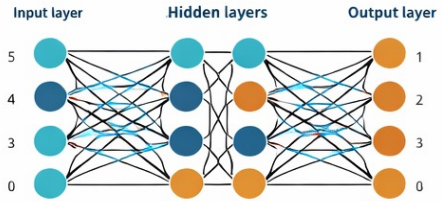
General ML World



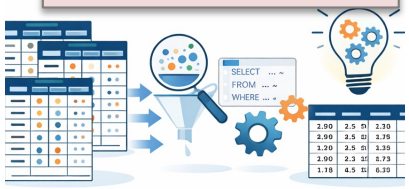


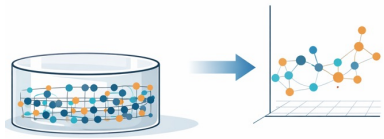


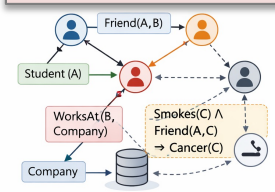




Feature generation	Low-dim embedding	Prob graphical models	Probabilistic programming	Deep neural networks
DB feature extraction	Tuple embedding	Markov Logic Networks	Generative queries	Hetero GNNs; rel. transform











Ex: ACORA | Ex: EmbDI | Ex: DeepDive | Ex: StarfishDB | Ex: KUMO

ML in the Relational-DB World

Databases: Tables + Graph of Records

Student

sid	name	started	address	gender	track
89	Anna	2022	Haifa	F	CS3
60	Bill	2024	Haifa	M	CS4
52	Carry	2023	Tel-Aviv	F	SE

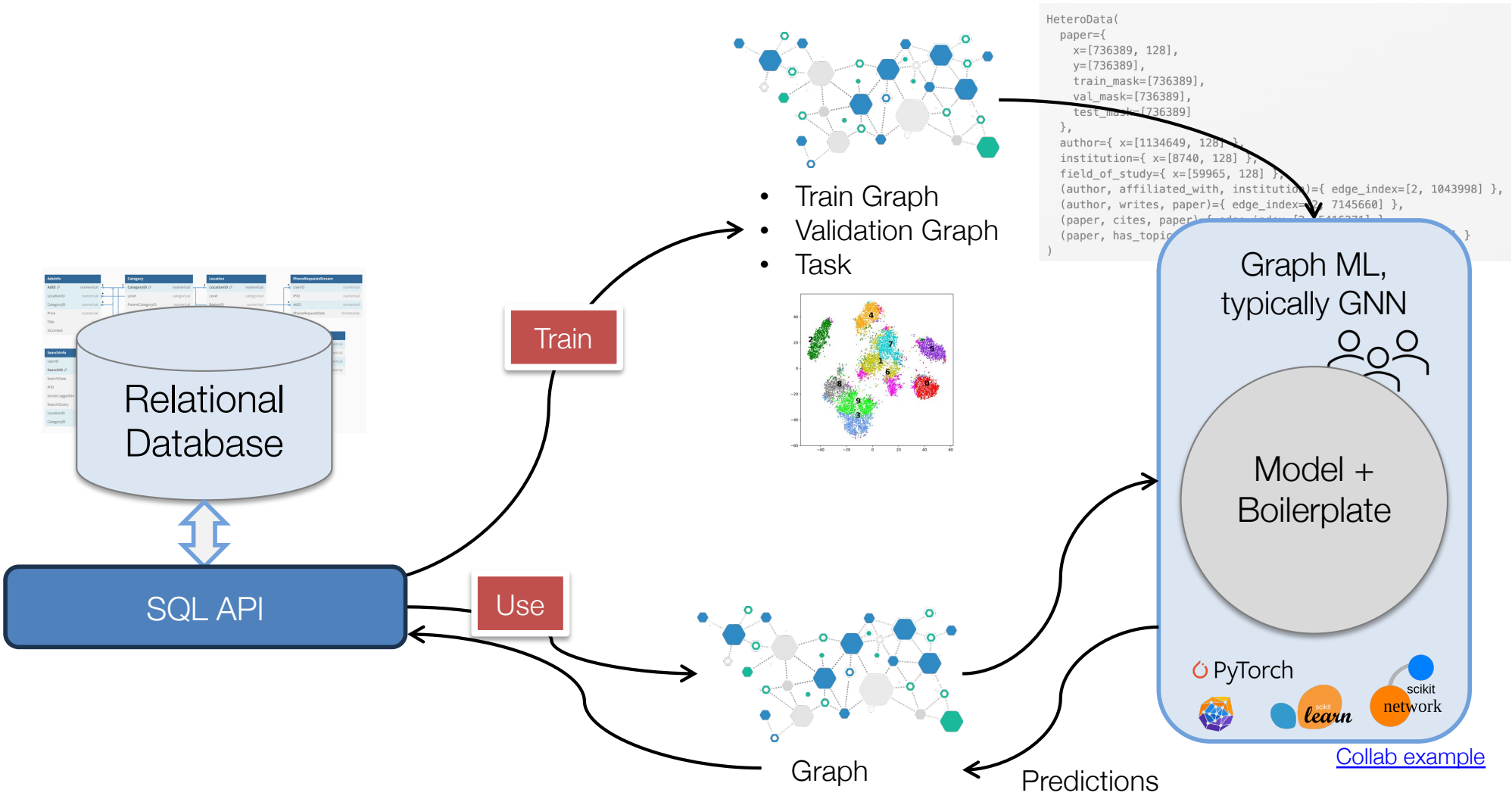
Take

sid	cid	year	semester
89	12	2025	1
89	25	2024	2
60	25	2024	2

Course

cid	name	credits	faculty
12	DB	3	CS
25	ML	3	CS
34	OS	4	ECS

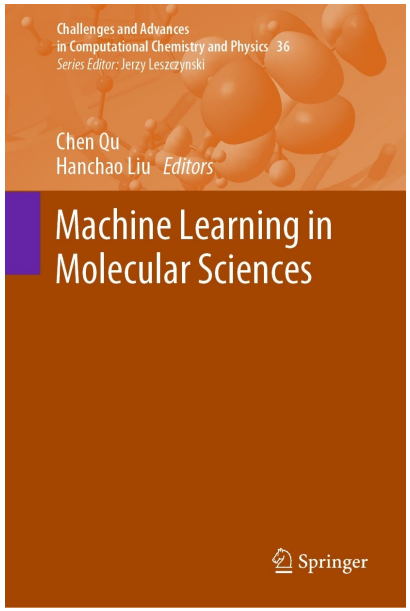
Common Workflow Nowadays

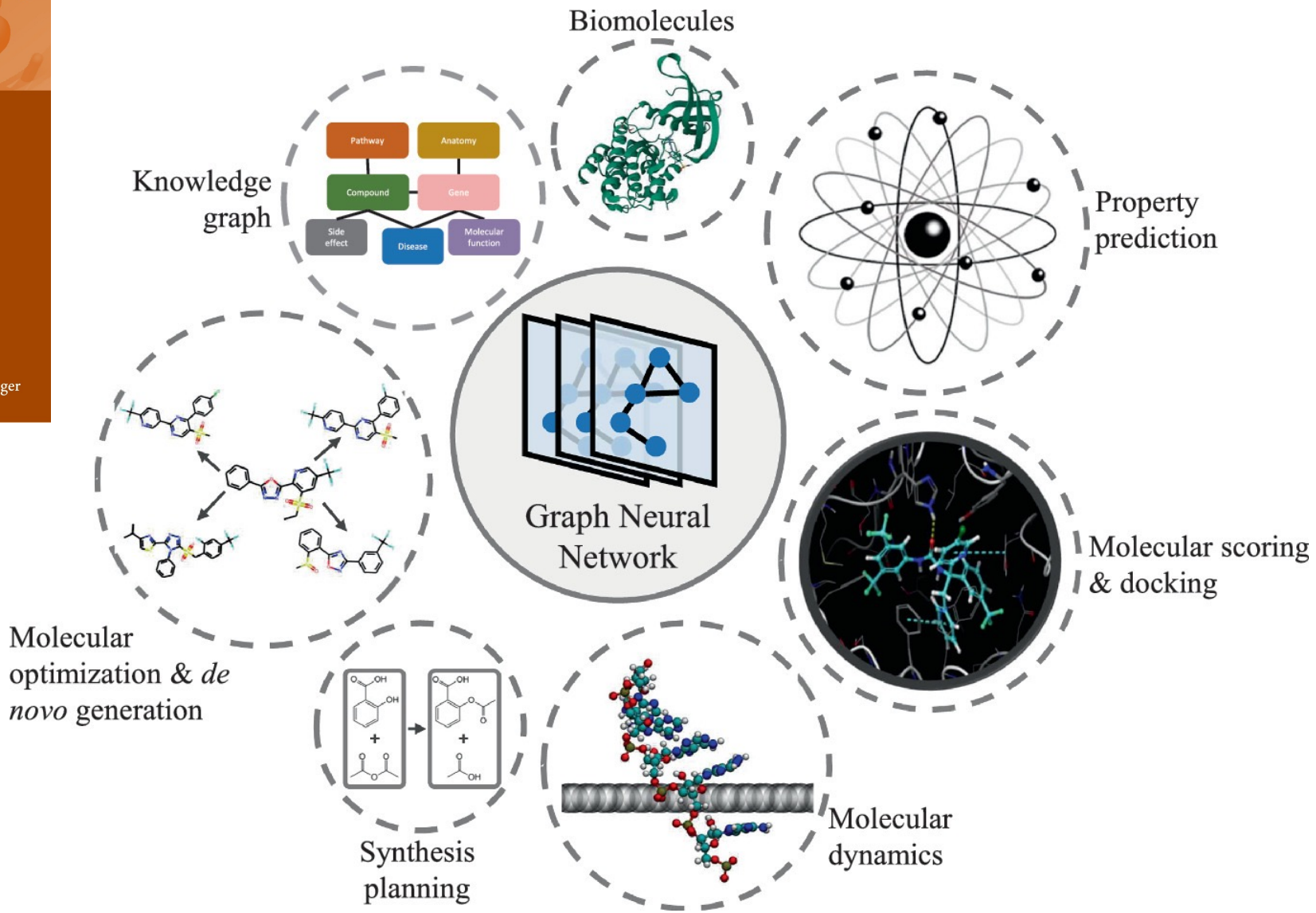


Graphs, Everywhere

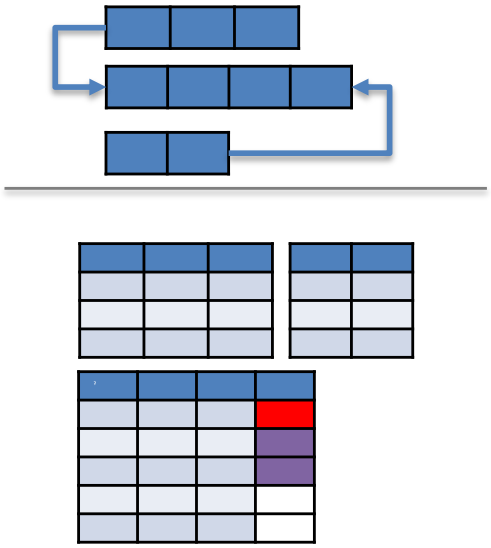
- Social networks
- Communication networks
- Molecules
- Internet pages
- Financial interactions
- Navigation and supply chains
- …

Graph Learning for Molecular Prediction

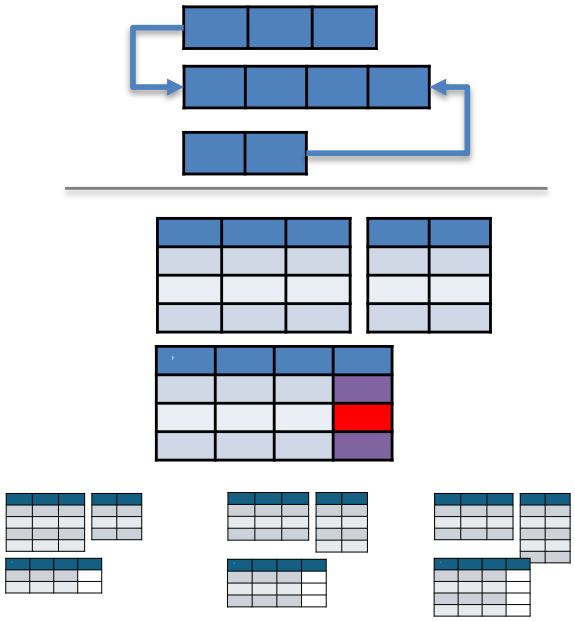




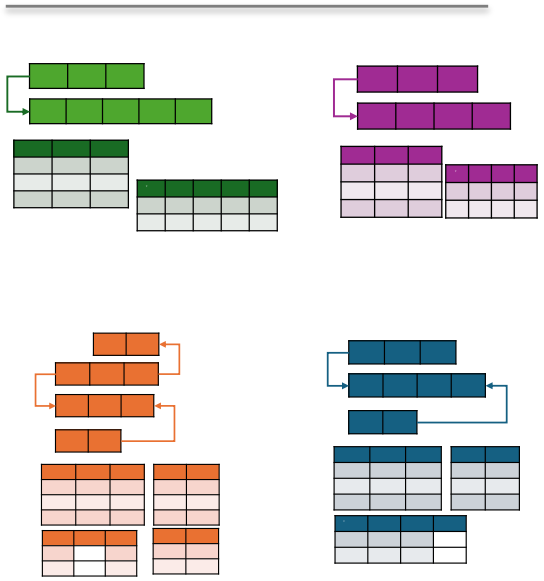
Transductive model:
Fixed DB & task

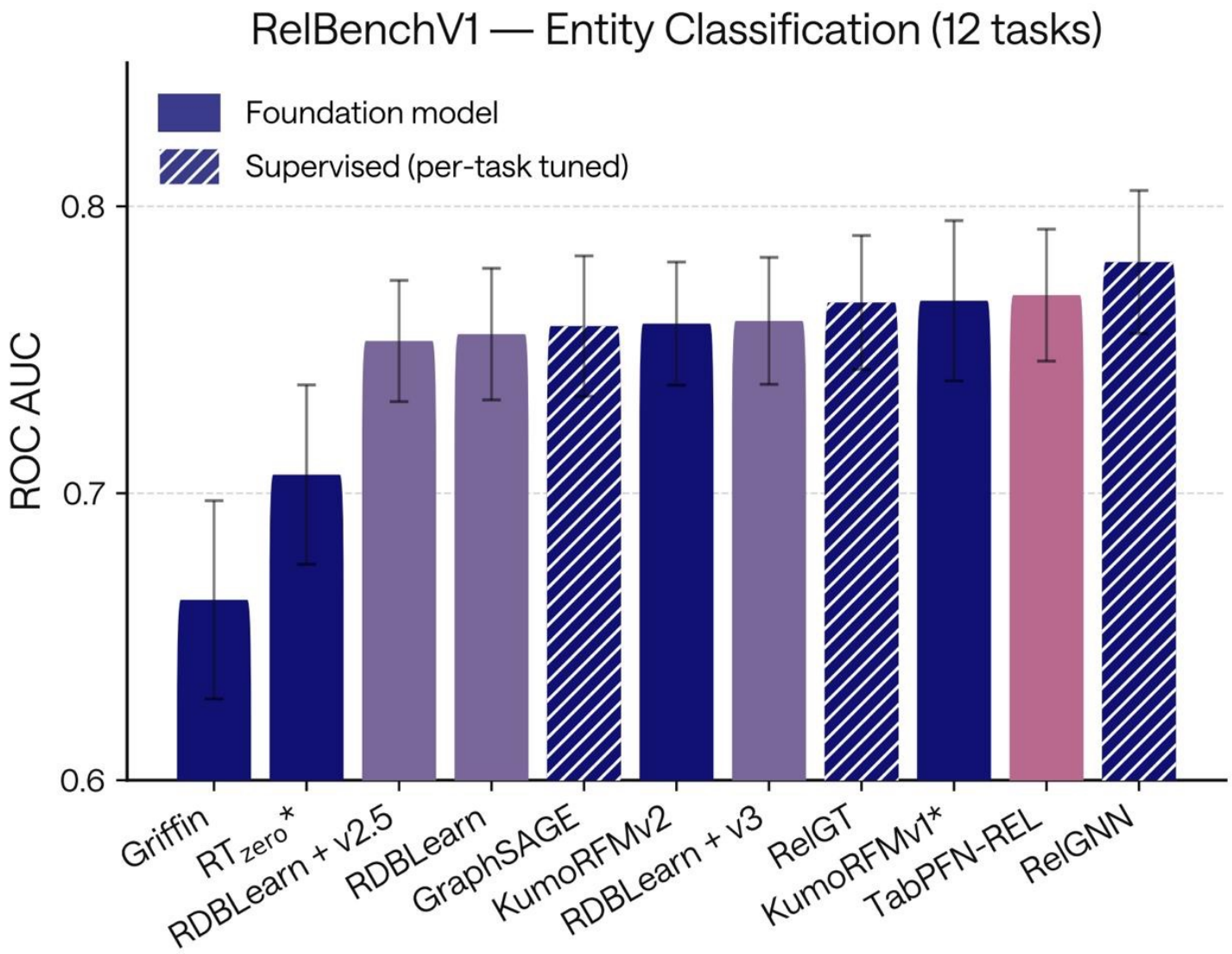


Inductive model:
Fixed schema & task, unknown DB



Foundation model:
Nothing known





Grinsztajn et al. TabPFN-3: Technical Report. CoRR abs/2605.13986 (2026)

Structured ML

Course Content

Core Modules in the Course

1. Basic **concepts**
   - Tables, Relations, Graphs, ML, Struct. ML
2. DB ML via **feature generation** & tabular ML
3. Low-dimensional **embedding** of structures
4. Overview of **deep learning**
   - Feedforward, perceptron, attention, transformers, back-propagation, GPUs/CUDA
5. DB ML via graph neural networks (**GNNs**)
6. Relational **foundation models**
7. Extra module on explainable structured ML
   - Boaz Berger

Good Luck!